# Operating Systems

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# Bilingual Course(双语课程)

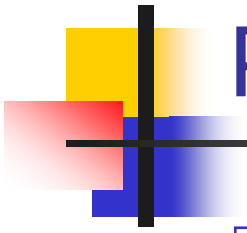
- The textbook is in English
- The lecture notes are in English
- The final exam paper will be in English

# Textbook

- William Stalling, Operating Systems：Internals and Design Principles(操作系统：精髓与设计原理) (8th Edition), Peason Education, Inc. 2017

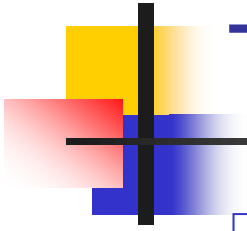
# Reference Books

- Andrew S. Tanenbaum, Operating Systems-Design and implementation (3rd Edition), Prentice Hall Press, 2006
- Maurice J. Bach, The Design of the Unix Operating System, Prentice Hall & CMP, 2000
- 于渊著. Orange's：一个操作系统的实现. 电子工业出版社. 2009



# Prerequisite

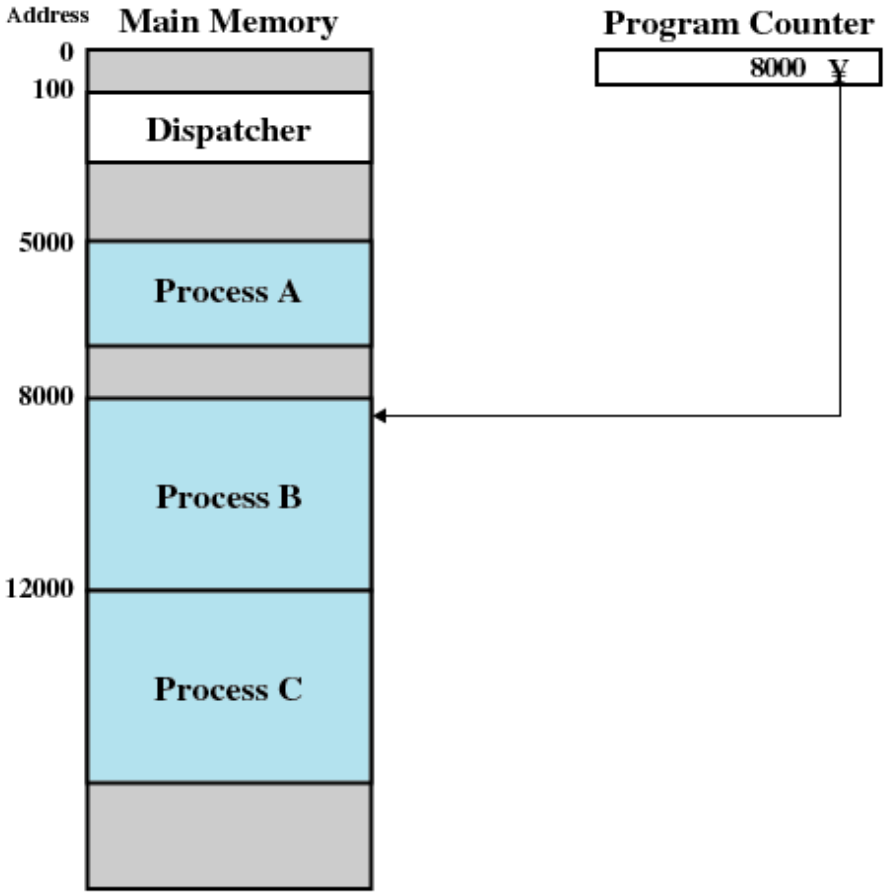
- Computer Principles
- Data Structures
- Programming Languages(C, Assembly language)



# Try to Answer the following Questions

- Why do many programs can be run in a computer with uniprocessor simultaneously?
- If we ran a program with bugs, other programs will be destroyed?
- How can we ran several instances of a program?
- Small main memory, big programs?
- How does a program save the running results ?
- Why does a program developed on one computer can be run on another computer?
- How does a computer manage a lot of devices ?

# Trace of Processes (1)



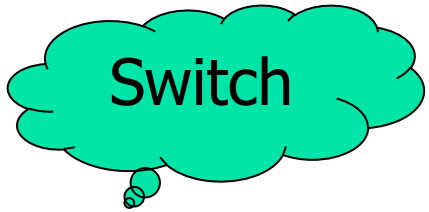
# Trace of Processes (2)

| | | |
|---|---|---|
| 5000 | 8000 | 12000 |
| 5001 | 8001 | 12001 |
| 5002 | 8002 | 12002 |
| 5003 | 8003 | 12003 |
| 5004 | | 12004 |
| 5005 | | 12005 |
| 5006 | | 12006 |
| 5007 | | 12007 |
| 5008 | | 12008 |
| 5009 | | 12009 |
| 5010 | | 12010 |
| 5011 | | 12011 |
| **(a) Trace of Process A** | **(b) Trace of Process B** | **(c) Trace of Process C** |

5000 = Starting address of program of Process A
8000 = Starting address of program of Process B
12000 = Starting address of program of Process C

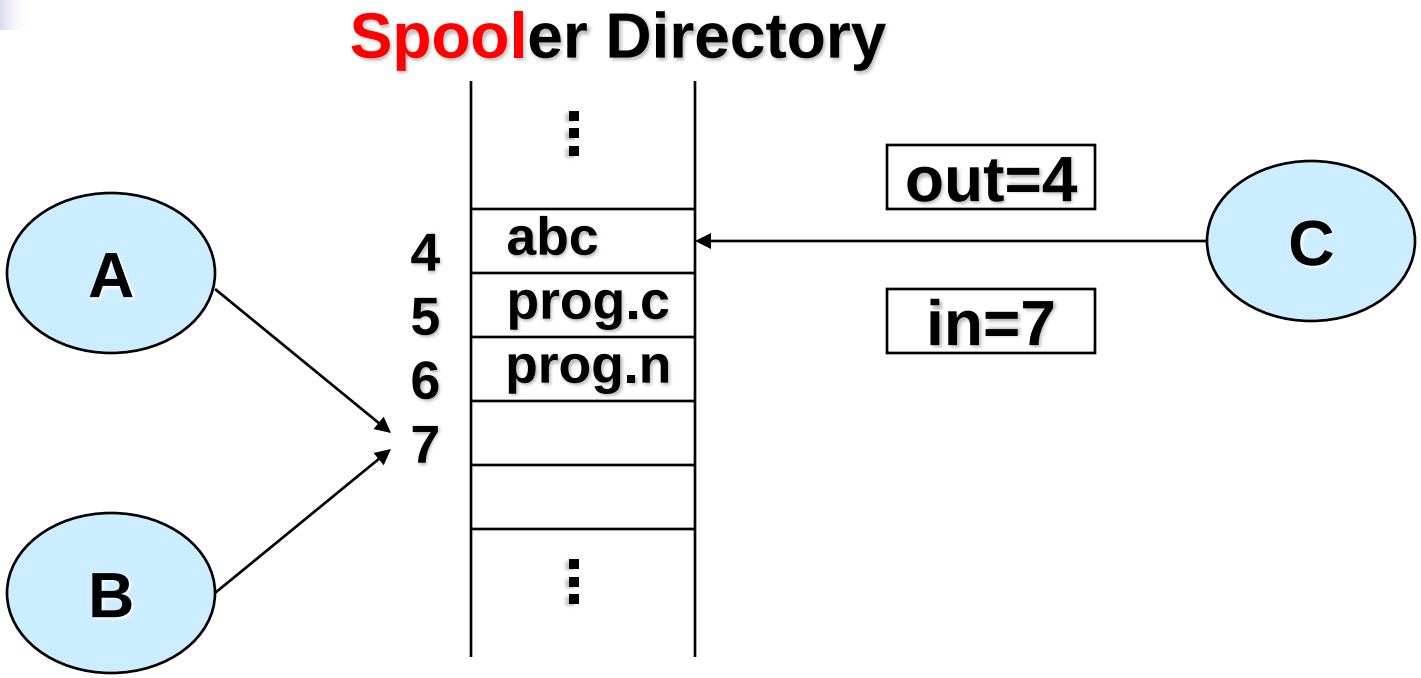
# Trace of Processes (3)

| | |
|---|---|
| 1 | 5000 |
| 2 | 5001 |
| 3 | 5002 |
| 4 | 5003 |
| 5 | 5004 |
| 6 | 5005 |
| ---------- | Time out |
| 7 | 100 |
| 8 | 101 |
| 9 | 102 |
| 10 | 103 |
| 11 | 104 |
| 12 | 105 |
| 13 | 8000 |
| 14 | 8001 |
| 15 | 8002 |
| 16 | 8003 |
| ---------- | I/O request |
| 17 | 100 |
| 18 | 101 |
| 19 | 102 |
| 20 | 103 |
| 21 | 104 |
| 22 | 105 |
| 23 | 12000 |
| 24 | 12001 |
| 25 | 12002 |
| 26 | 12003 |
| 27 | 12004 |
| 28 | 12005 |
| ---------- | Time out |
| 29 | 100 |
| 30 | 101 |
| 31 | 102 |
| 32 | 103 |
| 33 | 104 |
| 34 | 105 |
| 35 | 5006 |
| 36 | 5007 |
| 37 | 5008 |
| 38 | 5009 |
| 39 | 5010 |
| 40 | 5011 |
| ---------- | Time out |
| 41 | 100 |
| 42 | 101 |
| 43 | 102 |
| 44 | 103 |
| 45 | 104 |
| 46 | 105 |
| 47 | 12006 |
| 48 | 12007 |
| 49 | 12008 |
| 50 | 12009 |
| 51 | 12010 |
| 52 | 12011 |
| ---------- | Time out |






# Shared Memory Access (1)



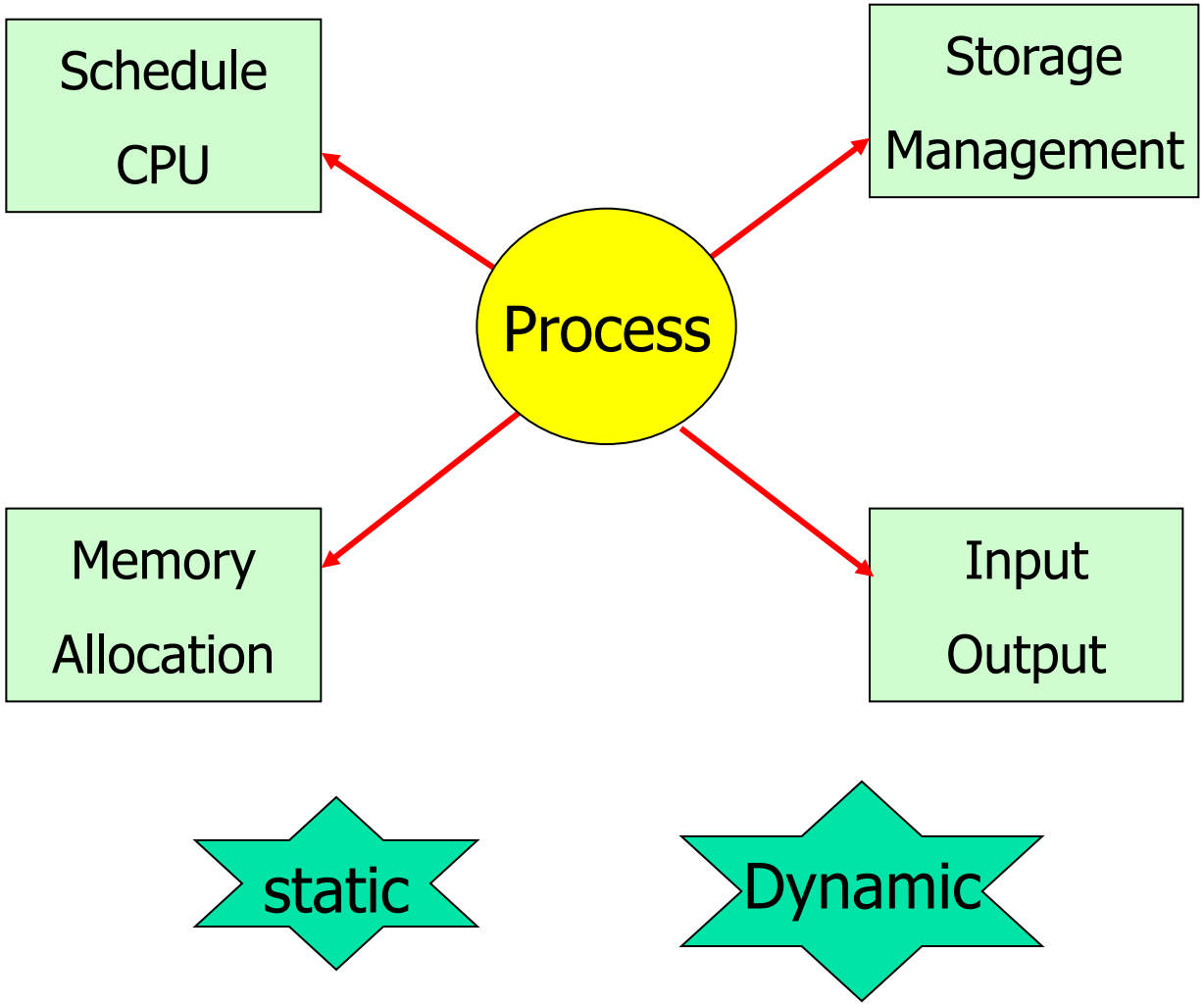

- out - next file will be printed
- in - next free slot
- c - printer daemon(打印机守护进程)

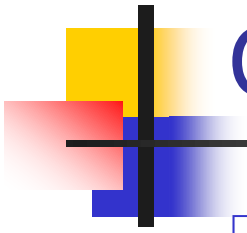
# Shared Memory Access (2)

- Process A read in=7, save in next_free_slot, clock interrupt occurs, process A is been switched
- Process B is scheduled to run, B read in=7, save the file to slot 7, update in=8, and continue to run
- Process A is restored to run, it checks next_free_slot=7, and save the file to slot 7, next_free_slot +1=>in, then in=8。
- The file which process B saved in slot 7 is lost，and printer daemon would not find this error

Serial(串行) and Parallel(并行)
Concurrent(并发)

# Program(程序) and Process(进程)





# Contents

- Process description and control
- Multithreading(多线程)
- Concurrency – synchronization(同步) and mutual exclusion(互斥)
- Concurrency – deadlock(死锁)
- Memory management
- Processor scheduling(调度)
- I/O management
- File systems